Affecting Corpora:Experiments with Automatic Affect Annotation System – A Case Study of the *2channel* Forum –

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Abstract

In this paper we present a system for automatic annotation of emotive information on corpora and use it to perform experiments investigating the functions of emotive expressions in conversation. The annotation system is based on a linguistic approach to emotions. It first separates emotive utterances from neutral and in the emotive ones seeks for expressions of specific emotion types. We performed two experiments with the system. In the first one we evaluated the system's usability. In the second one we annotated a large corpus of conversations from a popular Japanese Internet forum *2channel* to investigate the statistics and the communicative functions of emotive utterances.

1 Introduction

Research on linguistic aspects of expressing emotions is still a young field, however quickly developing and gaining interest of researchers. As for some remarkable examples, Wierzbicka (1999) studied cognitive linguistics of emotions among different cultures, Fussell and colleagues (2002) investigated the background theory of emotions, figurative language use, and social/cultural aspects of emotional communication, or Weigand and colleagues (2002) made an attempt to formulate a model of emotions in dialogic interactions.

Apart from the research generalizing about emotions, such as the above, there is also a wide range of study in the specific expressions of emotions. For example, Nakamura (1993) gathered a large collection of expressions describing states of emotions for the Japanese language, Baba (2003) studied Japanese mimetics in spoken discourse, Ono (2002) studied emphatic functions of Japanese particle *-da*[1] or Sasai (2006) examined *nanto*-type exclamatory sentences.

However, there is not much research on semiotic structures and communicative and pragmatic functions of expressions of emotions, and the less on these phenomena in the Japanese language. This is caused by several factors. Firstly, researchers engaged in linguistic studies on emotions usually perform the annotation manually. For example, Wilson and Wiebe (2005) started a long term project of manual annotation of English corpora. Annotating corpora manually greatly limits the scope of such research because of the workload and time costs. This is the reason much research in this field still depends on small corpora annotated manually (Minato et al., 2006). This problem could be solved by using a system for automatic analysis of corpora with emotive information. However, although recent years have brought research on emotions into focus in Computer Science (Picard 1997), till now there have not been developed any reliable system for emotive annotation of corpora. The systems designed using usual NLP methods (Tsuchiya et al., 2007; Tokuhisa et al., 2008) do not perform any linguistic analysis of utterance contents, which makes them incapable of distinguishing between emotive and non-emotive sentences and therefore inapplicable in linguistic research on emotions.

Therefore, in this paper we present a system for automatic annotation of corpora with emotive information. The system analyzes utterance contents from a linguistic point of view on expressing emotions. It first distinguishes if an utterance is emotive or not and then describes the semiotic structure of emotive utterances. We put the system into practice by performing an annotation of a large corpora to investigate the statistics and functions of emotive expressions in conversation.

[1] Italics are used to indicate expressions in Japanese.

2 The Emotive Function of Language

Linguistic means used in conversation to inform interlocutors of emotional states are described by the emotive function of language (Jakobson, 1960). Ptaszynski (2006) distinguished two kinds of its realizations in Japanese. The first one are emotive elements (or **emotems**) which indicate that emotions have been conveyed, but not detailing their specificity. This group is linguistically realized by interjections, exclamations, mimetic expressions, or vulgar language. The second are **emotive expressions** - parts of speech like nouns, verbs, adjectives or metaphors describing affective states. Nakamura (1993) classified emotive expressions into 10 emotion types said to be the most appropriate for the Japanese language. They can be translated as: joy, anger, gloom/sadness, fear, shame/shyness, fondness, dislike, excitement, relief and surprise (see Fig. 1).

Examples of sentences containing emotems and/or emotive expressions are shown in Table 1. Examples (1) and (2) are emotive sentences. (1) is an exclamative sentence, which is determined by the use of exclamative constructions *nante* (how/such a) and *nanda!* (exclamative sentence ending), and contains an emotive expression *kimochi ii* (to feel good). (2) is also an exclamative. It is easily recognizable by the use of an interjection *iyaa*, an adjective in the function of interjection *sugoi* (great), and by the emphatic particle *-ne*. However, it does not contain any emotive expressions and therefore it is ambiguous whether the emotions conveyed by the speaker are positive or negative. The examples (3) and (4) show non emotive sentences. (3), although containing an emotive verb *aishiteiru* (to love), is a generic statement and, if not put in a specific context, does not convey any emotions. Finally, (4) is a simple declarative sentence without any emotive value.

3 Defining Emotive Linguistic Features

We gathered databases of emotems and emotive expressions according to Ptaszynski's two-part classification. The feature set was defined in a way similar to the one proposed by Alm et al. (2005), by using multiple features to handle emotive sentences. Alm however, designed their research for English children's stories, whereas we focus on utterances in Japanese, and therefore used Ptaszynski's classification as more appropriate for our research.

Table 1: Examples of sentences containing emotems (underlined) and/or emotive expressions (bold type font).

Example of a sentence (English translation)	emotems	**emotive expressions**
(1) *Kyo wa nante* ***kimochi ii*** *hi nanda!* (Today is such a nice day!)	yes	yes
(2) *Iyaa, sore wa sugoi desu ne!* (Whoa, that's great!)	yes	no
(3) *Ryoushin wa minna jibun no kodomo wo* ***aishiteiru***. (All parents love their children.)	no	yes
(4) *Kore wa hon desu.* (This is a book.)	no	no

3.1 Emotems

Into the group of emotive elements, formally visualisable as textual representations of speech, Ptaszynski (2006) includes the following lexical and syntactical structures.

Exclamative utterance. We agree with Beijer's (2002) definition of exclamative/emotive utterances, as every utterance in which the speaker is emotionally involved, and this involvement is expressed linguistically. The research on exclamatives in Japanese (Ono, 2002; Sasai, 2006) provides a wide scope of topics useful as features for our system. Some of the exclamative structures are: *nan(te/to/ka)–*, *–darou*, or *–da(yo/ne)*, partially corresponding to wh-exclamatives in English (see the first sentence in Table 1).

Interjections are typical emotems. Interjections in Japanese are, e.g. *waa*, *yare-yare* or *iyaa*. See sentence (2) in Table 1.

Casual Speech. Casual speech is not an emotem per se, however, many structures of casual speech are used when expressing emotions. Examples of casual language use are modifications of adjective and verb endings *-ai* to *-ee*, like in the example: *Ha ga itee!* (My tooth hurts!), or abbreviations of forms *-noda* into *-nda*, like in the example: *Nani yattenda yo!?* (What the hell are you doing!?).

***Gitaigo*.** Baba (2003) distinguishes *gitaigo* (mimetic expressions) as emotems specific for the Japanese language. Not all mimetics are emotive, but rather they can be classified into emotive mimetics (describing one's emotions), and sensation/state mimetics (describing manner and appearance). Examples of emotive *gitaigo* are: *iraira* (be irritated), or *hiyahiya* (be in fear, nervous), like in the sentence: *Juugeki demo sareru n janai ka to omotte, hiyahiya shita ze.* (I thought he was gonna shoot me - I was petrified.)

Emotive marks. This group contains punctuation marks used as a textual representations of emotive

intonation features. The most obvious example is exclamation mark "!". In Japanese, marks like ellipsis "...", prolongation marks, like "–", or "∼", are also used to inform interlocutors that emotions have been conveyed (see Table 1).

Hypocoristics (endearments) in Japanese express emotions and attitudes towards an object by the use of diminutive forms of a name or status of the object (*Ai* [girl's name] vs *Ai-chan* [/endearment/]; *o-nee-san* [older sister] vs *o-nee-chan* [sis]). Example: *Saikin Oo-chan to Mit-chan ga bokura to karamu youni nattekita!!* (Oo-chan and Mit-chan has been palling around with us lately!!)

Vulgarisms. The use of vulgarisms usually accompanies expressing emotions. However, despite a general belief that vulgarisms express only negative meaning, Ptaszynski (2006) noticed that they can be also used as expressions of strong positive feelings, and Sjöbergh (2006) showed, that they can also be funny, when used in jokes, like in the example: *Mono wa mono dakedo, fuete komarimasu mono wa nanda–? Bakamono.* (A thing (*mono*) is a thing, but what kind of thing is bothersome if they increase? Idiots (*bakamono*).)

3.2 Emotive Expressions

This group contains a lexicon of expressions describing emotional states. Some examples are:
adjectives: *ureshii* (happy), *sabishii* (sad);
nouns: *aijou* (love), *kyofu* (fear);
verbs: *yorokobu* (to feel happy), *aisuru* (to love);
fixed phrases/idioms: *mushizu ga hashiru* (give one the creeps [of hate]), *kokoro ga odoru* (one's heart is dancing [of joy]);
proverbs: *dohatsuten wo tsuku* (be in a towering rage), *ashi wo fumu tokoro wo shirazu* (be with one's heart up the sky [of happiness]);
metaphors/similes: *itai hodo kanashii* (sadness like a physical pain), *aijou wa eien no honoo da* (love is an eternal flame);
Such a lexicon can be used to express emotions, like in the first example in the Table 1, however, it can also be used to formulate non-emotive declarative sentences (third example in Table 1).

4 ML-Ask: Affect Annotation System

Based on the linguistic approach towards emotions we constructed ML-Ask (eMotive eLements–SeeK & Analyse) system for automatic annotation of corpora with emotive information. The emotems databases for the system were gathered manually from other research and grouped into five types (code, reference research and number of gathered items in square, round and curly brackets, respectively):
1.**[EX]** Interjections and structures of exclamative and emotive-casual utterances (Nakamura, 1993; Oshima-Takane et al., 1995-1998; Tsuchiya, 1999; Ono, 2002). {477}
2.**[GI]** *Gitaigo* (Nakamura, 1993; Oshima-Takane et al., 1995-1998; Baba, 2003). {213}
3.**[HY]** Hypocorystrics (Kamei et al., 1996). {8}
4.**[VU]** Vulgarisms (Sjöbergh, 2008). {200}
5.**[EM]** Emotive marks (Kamei et al., 1996). {9}

These databases were used as a core for ML-Ask. Nakamura's (1993) dictionary functions as a database of emotive expressions (code: **[EMO-X]**, 2100 items in total). The breakdown with number of items per emotion type was as follows: joy {224}, anger {199}, gloom {232}, fear {147}, shame {65}, fondness {197}, dislike {532}, excitement {269}, relief {106}, surprise {129}.

A textual input utterance is matched to the emotem databases and emotive information is annotated. The system first determines whether an utterance is emotive (appearance of at least one emotive feature), extracts all features from the sentence and describes the structure of the emotive utterance. This is the system's main procedure for emotive information annotation of large corpora. Next, in all utterances determined as emotive, the system searches for emotive expressions from the databases (Nakamura's collection). This procedure is used to verify: **1)** How many of all determined emotive utterances contain emotive expressions; **2)** Whether the system is capable of annotating specific emotion types in conversation? Examples of analysis are shown in Table 2.

However, keyword-based extraction allowed mismatching the specific emotion types. To avoid this, we applied Contextual Valence Shifters.

4.1 Contextual Valence Shifters in ML-Ask

The idea of Contextual Valence Shifters (CVS) was proposed by Polanyi and Zaenen (2004), who distinguished 2 kinds of CVS: negations (changing the valence polarity of an evaluative word) and intensifiers (intensifying its semantic orientation). Examples of negations in Japanese are: *amari -nai* (not quite-), *-to wa ienai* (cannot say it is-), or *-te wa ikenai* (don't-). Intensifiers are: *sugoku-* (-a lot) or *kiwamete-* (extremely).

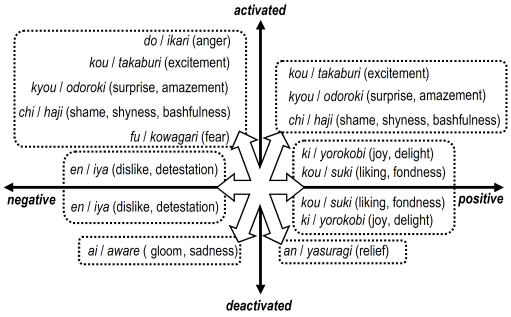


Figure 1: Nakamura's emotions on Russell's space.

When a CVS structure is discovered, ML-Ask changes the valence polarity of the detected emotion. To specify the emotion types afterwards, we applied the 2-dimensional model of affect (Russell, 1980) which assumes that all emotions can be described in 2 dimensions: valence polarity (positive/negative) and activation (activated/deactivated). Nakamura's emotion types were mapped on this model and their affiliation to one of the spaces was determined. After valence shifting the emotion type is determined as the one with valence polarity and activation parameters opposite to the contrasted emotion (see Fig. 1 and (5) in Table 2).

5 Evaluation

In the evaluation we contrasted the linguistic approach to emotive information annotation (represented by ML-Ask) with the laypeople's one. We gathered a corpus of natural utterances through an anonymous survey in which we asked people of different ages and social groups to remember a conversation with a friend and write three sentences from the conversation: 1) free, 2) emotive, and 3) non-emotive. From this collection we used only the 80 utterances meant to be either emotive (40 items) or non-emotive (40 items). The participants also annotated the specific emotion types on emotive utterances.

Since laypeople, not possessing any linguistic knowledge, are unable to describe the structure of emotive utterances, we checked whether ML-Ask could distinguish between emotive and non-emotive utterances. As a result, ML-Ask annotated correctly 72 from 80 utterances (90%). In 2 cases the system wrongly annotated utterances as "emotive", in 6 cases it was the opposite. The system's agreement with annotators was indicated as very high (kappa=.8). Therefore, ML-Ask proved its reliability in identifying emotive utterances

Table 2: 4 examples of the ML-Ask analysis. From top line: example in Japanese, emotive information annotation, English translation. Emotems-underlined, emotive expressions-bold type font, n-noun, ptl-particle, AUX-auxiliary verb.

(1)	*Kyo*	*wa*	*nante*	***kimochi ii***	*hi*	*nanda*	*!*
	noun	ptl:THEM	EX:nante	**EMO-X:joy**	noun:SUBJ	EX:nanda	EM:!
(2)	*Iyaa,*	*sore*	*wa*	*sugoi*	*desu*	*ne*	*!*
	EX:iyaa	noun	ptl:THEM	EX:sugoi	AUX	EX:ne	EM:!
(5)	***Akirame***		*cha*	*ikenai*		*yo*	*!*
	EMO-X:dislike		EX:cha	CVS:cha-ikenai{→joy}		EX:yo	EM:!

Translation: Don't cha give up!

We also checked the ratio of emotive utterances containing emotive expressions to verify reliability of Nakamura's dictionary. The system extracted emotive expressions from 9 out of 40 emotive sentences (22.5%). All were extracted correctly, which proved basic reliability of Nakamura's collection. However, the coverage is not high, which has two reasons: 1) Nakamura stopped updating his dictionary in 1993 and the lexicon is out-of-date (only 2100 expressions); 2) instead of using straight forward emotive expressions, people would rather use ambiguous emotive utterances in which emphasis is based on the context (see example (2)). However, such sentences can be handled using NLP methods for Affect Recognition (Tokuhisa et al., 2008).

6 Corpus Annotation Experiment

After confirming the system's usability we performed an annotation of a large corpus of Internet forum discussions to verify Ptaszynski's (2006) thesis saying that emotive utterances fulfill in conversation also other functions than expressing emotions and attitudes.

6.1 *2channel* Forum as a Corpus

As the corpus we used a collection of discussions from a popular Japanese forum *2channel* [2], where every day millions of people discuss the current topics. The special feature of the forum is the anonymity of speech. Utterances without signature appear on the board as *Nanashi-san* (Mr. Nameless). As Ptaszynski argues, such restrictions force users to create communication strategies compensating for the limitations of the medium. One kind of such strategy is a frequent usage of emotive expressions. Matsumura et al.

[2] http://www.2ch.net/

(2004) go further and argue that expressing negative emotions helps keeping the discussion up.

To verify these theories we used a collection of discussions that took place on the forum as a corpus for annotations. The collection was officially published in Japan as an experimental novel under the title *Densha otoko* (Train man) (Nakano, 2005), and contains 177,553 characters in 1,840 utterances, divided into 6 parts. On this corpus Ptaszynski (2006), with a help of a Japanese native speaker, annotated manually the emotive utterances. However, he focused only on the ones containing emotive expressions, since the ambiguously emotive utterances appeared too frequently. We annotated this corpus using ML-Ask and compared the results.

6.2 Annotation Results and Discussion

First we used the system to calculate the concordance of emotive utterances in the corpus. Since the system proved its high reliability in identifying emotive utterances, we used its results as the correct ones. The result was 1506 emotive utterances (81% of all corpus). It is reasonable, knowing the forum's reputation. The number also confirms Ptaszynski's thesis that emotive expressions are used on *2channel* frequently as a pragmatic mean to sustain and support the discussion.

Next, we calculated how many of the emotive utterances contained specified emotive expressions and compared the results to manual annotations. For different parts of the corpora the system annotated from 19% to 25% of the whole contents of a corpora part as emotive utterances. This corresponds to the scope of 40% to 75% (Average of 58%) of the human annotations, which we consider as an accuracy ratio (see Figure 2-1)). The variation in ratios is caused by the lacks in the databases described in section **5**. The results were considered to be very statistically significant with P value = .0052 and, when contrasted, showed similar tendency - emotive expressions appeared in the largest number in the last part of the corpus.

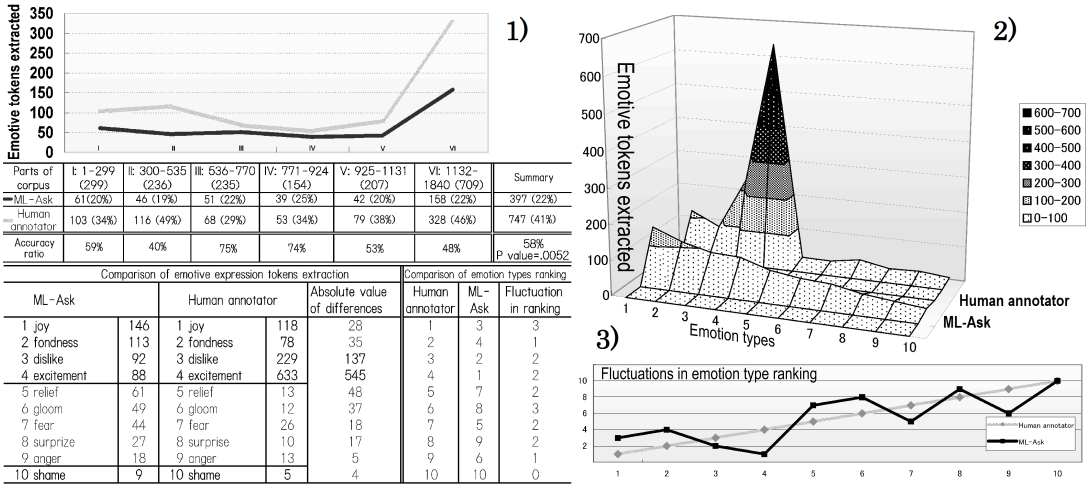
The similarities appeared also in the emotion type annotations. Although the overall number of the emotive tokens extracted by the system and annotated manually differed, the detailed analysis revealed that the system had problems only with determining about 'dislike' and 'excitement' (see Figure 2-2)). This means that to express these two emotions in particular *2channel* users mostly use context dependent ambiguous emotive expressions. Except of these two emotion types the system could extract other emotive tokens similarly to human annotator. There were 90% of agreements observed with the strength o agreement coefficient considered to be good (Kappa = .681) and very statistically significant (P value = .0035).

Finally, we checked the tendency in providing the ranking of all emotion types appearing in the corpus. The ranking was based on the number of tokens extracted. The system's ranking was compared to human annotations considered as gold standard. Assigning the ranking places was scored from 10 points (a perfect hit, ranking place fluctuation = 0) to 0 (ranking place for an emotion type not assigned at all, fluctuation = 10). A maximum number of points to be gathered this way was 100 (all ranking places perfectly assigned), minimum was 0 points (no emotion type assigned). ML-Ask's fluctuations were from 0 (for 'shame') to 3 (for 'joy' and 'gloom') (see Figure 2-3)). With such results ML-Ask acquired a high score of 82 points. The differences in ranking places fluctuations and therefore the score in emotion type ranking assessment was considered extremely statistically significant by conventional criteria (P value = .0002). The general view on the results revealed that the group including the majority of tokens was ranked more similarly (emotion types in bold type font in the table on Figure 2).

7 Conclusions and Future work

In this paper we presented ML-Ask, a system for automatic annotation of corpora with emotive information. A need for such system is expressed in many research on emotion processing. After verifying the system's utility we performed an annotation of a corpus containing discussions from a popular Japanese forum *2channel*. With manual annotation of the same corpus as gold standard the system resulted as sufficiently (and significantly) successful in providing information on tendencies of emotive utterances in conversations.

In the future we plan to improve the system by updating the emotive expressions lexicon and statistically disambiguate emotems by attaching to them potential emotive affiliations (e.g. an exclamation mark would be used with "excitement", rather than with "gloom"). We also plan to perform experiments including large scale annotations of other natural dialogue corpora to find out



Parts of corpus	I: 1-299 (299)	II: 300-535 (236)	III: 536-770 (235)	IV: 771-924 (154)	V: 925-1131 (207)	VI: 1132-1840 (709)	Summary
ML-Ask	61(20%)	46 (19%)	51 (22%)	39 (25%)	42 (20%)	158 (22%)	397 (22%)
Human annotator	103 (34%)	116 (49%)	68 (29%)	53 (34%)	79 (38%)	328 (46%)	747 (41%)
Accuracy ratio	59%	40%	75%	74%	53%	48%	58% P value=.0052

Comparison of emotive expression tokens extraction					Comparison of emotion types ranking		
ML-Ask		Human annotator		Absolute value of differences	Human annotator	ML-Ask	Fluctuation in ranking
1 joy	146	1 joy	118	28	1	3	3
2 fondness	113	2 fondness	78	35	2	4	1
3 dislike	92	3 dislike	229	137	3	2	2
4 excitement	88	4 excitement	633	545	4	1	2
5 relief	61	5 relief	13	48	5	7	2
6 gloom	49	6 gloom	12	37	6	8	3
7 fear	44	7 fear	26	18	7	5	2
8 surprize	27	8 surprise	10	17	8	9	2
9 anger	18	9 anger	13	5	9	6	1
10 shame	9	10 shame	5	4	10	10	0

Figure 2: Results of ML-Ask in comparison with human annotations.

about the functions emotive utterances play in different contexts and conversation environments.

Acknowledgments

This research was partially supported by the Nissan Science Foundation's Research Grant and The GCOE Program from Japan's Ministry of Education, Culture, Sports, Science and Technology.

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